

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

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<https://github.com/deepseek-ai/DeepSeek-Math>

Abstract

Mathematical reasoning poses a significant challenge for language models due to its complex and structured nature. In this paper, we introduce DeepSeekMath 7B, which continues pre-training DeepSeek-Coder-Base-v1.5 7B with 120B math-related tokens sourced from Common Crawl, together with natural language and code data. DeepSeekMath 7B has achieved an impressive score of 51.7% on the competition-level MATH benchmark without relying on external toolkits and voting techniques, approaching the performance level of Gemini-Ultra and GPT-4. Self-consistency over 64 samples from DeepSeekMath 7B achieves 60.9% on MATH. The mathematical reasoning capability of DeepSeekMath is attributed to two key factors: First, we harness the significant potential of publicly available web data through a meticulously engineered data selection pipeline. Second, we introduce Group Relative Policy Optimization (GRPO), a variant of Proximal Policy Optimization (PPO), that enhances mathematical reasoning abilities while concurrently optimizing the memory usage of PPO.

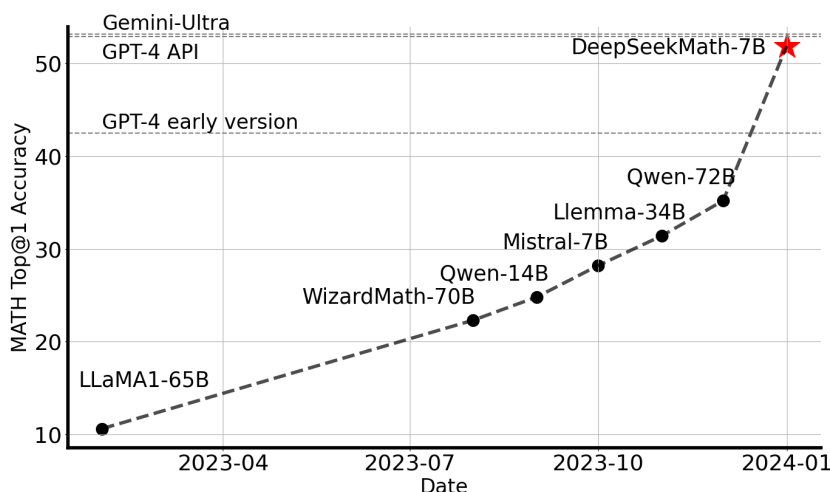


Figure 1 | Top1 accuracy of open-source models on the competition-level MATH benchmark (Hendrycks et al., 2021) without the use of external toolkits and voting techniques.

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† Work done during internship at DeepSeek-AI.

1. Introduction

Large language models (LLM) have revolutionized the approach to mathematical reasoning in artificial intelligence, spurring significant advancements in both the quantitative reasoning benchmark (Hendrycks et al., 2021) and the geometry reasoning benchmark (Trinh et al., 2024). Moreover, these models have proven instrumental in assisting humans in solving complex mathematical problems (Tao, 2023). However, cutting-edge models such as GPT-4 (OpenAI, 2023) and Gemini-Ultra (Anil et al., 2023) are not publicly available, and the currently accessible open-source models considerably trail behind in performance.

In this study, we introduce DeepSeekMath, a domain-specific language model that significantly outperforms the mathematical capabilities of open-source models and approaches the performance level of GPT-4 on academic benchmarks. To achieve this, we create the DeepSeekMath Corpus, a large-scale high-quality pre-training corpus comprising 120B math tokens. This dataset is extracted from the Common Crawl (CC) using a fastText-based classifier (Joulin et al., 2016). In the initial iteration, the classifier is trained using instances from OpenWebMath (Paster et al., 2023) as positive examples, while incorporating a diverse selection of other web pages to serve as negative examples. Subsequently, we employ the classifier to mine additional positive instances from the CC, which are further refined through human annotation. The classifier is then updated with this enhanced dataset to improve its performance. The evaluation results indicate that the large-scale corpus is of high quality, as our base model DeepSeekMath-Base 7B achieves 64.2% on GSM8K (Cobbe et al., 2021) and 36.2% on the competition-level MATH dataset (Hendrycks et al., 2021), outperforming Minerva 540B (Lewkowycz et al., 2022a). In addition, the DeepSeekMath Corpus is multilingual, so we notice an improvement in Chinese mathematical benchmarks (Wei et al., 2023; Zhong et al., 2023). We believe that our experience in mathematical data processing is a starting point for the research community, and there is significant room for improvement in the future.

DeepSeekMath-Base is initialized with DeepSeek-Coder-Base-v1.5 7B (Guo et al., 2024), as we notice that starting from a code training model is a better choice compared to a general LLM. Furthermore, we observe the math training also improves model capability on MMLU (Hendrycks et al., 2020) and BBH benchmarks (Suzgun et al., 2022), indicating it does not only enhance the model’s mathematical abilities but also amplifies general reasoning capabilities.

After pre-training, we apply mathematical instruction tuning to DeepSeekMath-Base with chain-of-thought (Wei et al., 2022), program-of-thought (Chen et al., 2022; Gao et al., 2023), and tool-integrated reasoning (Gou et al., 2023) data. The resulting model DeepSeekMath-Instruct 7B beats all 7B counterparts and is comparable with 70B open-source instruction-tuned models.

Furthermore, we introduce the Group Relative Policy Optimization (GRPO), a variant reinforcement learning (RL) algorithm of Proximal Policy Optimization (PPO) (Schulman et al., 2017). GRPO foregoes the critic model, instead estimating the baseline from group scores, significantly reducing training resources. By solely using a subset of English instruction tuning data, GRPO obtains a substantial improvement over the strong DeepSeekMath-Instruct, including both in-domain (GSM8K: 82.9% $\rightarrow$ 88.2%, MATH: 46.8% $\rightarrow$ 51.7%) and out-of-domain mathematical tasks (e.g., CMATH: 84.6% $\rightarrow$ 88.8%) during the reinforcement learning phase. We also provide a unified paradigm to understand different methods, such as Rejection Sampling Fine-Tuning (RFT) (Yuan et al., 2023a), Direct Preference Optimization (DPO) (Rafailov et al., 2023), PPO and GRPO. Based on such a unified paradigm, we find that all these methods are conceptualized as either direct or simplified RL techniques. We also conduct extensive experiments, e.g., online v.s. offline training, outcome v.s. process supervision, single-turn v.s. iterative RL and so on,

to deeply investigate the essential elements of this paradigm. At last, we explain why our RL boosts the performance of instruction-tuned models, and further summarize potential directions to achieve more effective RL based on this unified paradigm.

1.1. Contributions

Our contribution includes scalable math pre-training, along with the exploration and analysis of reinforcement learning.

Math Pre-Training at Scale

- Our research provides compelling evidence that the publicly accessible Common Crawl data contains valuable information for mathematical purposes. By implementing a meticulously designed data selection pipeline, we successfully construct the DeepSeekMath Corpus, a high-quality dataset of 120B tokens from web pages filtered for mathematical content, which is almost 7 times the size of the math web pages used by Minerva (Lewkowycz et al., 2022a) and 9 times the size of the recently released OpenWebMath (Paster et al., 2023).
- Our pre-trained base model DeepSeekMath-Base 7B achieves comparable performance with Minerva 540B (Lewkowycz et al., 2022a), indicating the number of parameters is not the only key factor in mathematical reasoning capability. A smaller model pre-trained on high-quality data could achieve strong performance as well.
- We share our findings from math training experiments. Code training prior to math training improves models’ ability to solve mathematical problems both with and without tool use. This offers a partial answer to the long-standing question: *does code training improve reasoning abilities?* We believe it does, at least for mathematical reasoning.
- Although training on arXiv papers is common, especially in many math-related papers, it brings no notable improvements on all mathematical benchmarks adopted in this paper.

Exploration and Analysis of Reinforcement Learning

- We introduce Group Relative Policy Optimization (GRPO), an efficient and effective reinforcement learning algorithm. GRPO foregoes the critic model, instead estimating the baseline from group scores, significantly reducing training resources compared to Proximal Policy Optimization (PPO).
- We demonstrate that GRPO significantly enhances the performance of our instruction-tuned model DeepSeekMath-Instruct, by solely using the instruction-tuning data. Furthermore, we observe enhancements in the out-of-domain performance during the reinforcement learning process.
- We provide a unified paradigm to understand different methods, such as RFT, DPO, PPO, and GRPO. We also conduct extensive experiments, e.g., online v.s. offline training, outcome v.s. process supervision, single-turn v.s. iterative reinforcement learning, and so on to deeply investigate the essential elements of this paradigm.
- Based on our unified paradigm, we explore the reasons behind the effectiveness of reinforcement learning, and summarize several potential directions to achieve more effective reinforcement learning of LLMs.

1.2. Summary of Evaluations and Metrics

- **English and Chinese Mathematical Reasoning:** We conduct comprehensive assessments of our models on English and Chinese benchmarks, covering mathematical problems

A.1.4. Direct Preference Optimization (DPO)

The objective of DPO is:

$$\mathcal{J}_{DPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o^+, o^- \sim \pi_{sft}(O|q)] \log \sigma \left(\beta \frac{1}{|o^+|} \sum_{t=1}^{|o^+|} \log \frac{\pi_{\theta}(o_t^+|q, o_{<t}^+)}{\pi_{ref}(o_t^+|q, o_{<t}^+)} - \beta \frac{1}{|o^-|} \sum_{t=1}^{|o^-|} \log \frac{\pi_{\theta}(o_t^-|q, o_{<t}^-)}{\pi_{ref}(o_t^-|q, o_{<t}^-)} \right) \quad (12)$$

The gradient of $\mathcal{J}_{DPO}(\theta)$ is:

$$\begin{aligned} \nabla_{\theta} \mathcal{J}_{DPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o^+, o^- \sim \pi_{sft}(O|q)] & \left(\frac{1}{|o^+|} \sum_{t=1}^{|o^+|} GC_{DPO}(q, o, t) \nabla_{\theta} \log \pi_{\theta}(o_t^+|q, o_{<t}^+) \right. \\ & \left. - \frac{1}{|o^-|} \sum_{t=1}^{|o^-|} GC_{DPO}(q, o, t) \nabla_{\theta} \log \pi_{\theta}(o_t^-|q, o_{<t}^-) \right) \end{aligned} \quad (13)$$

Data Source: question in SFT dataset with outputs sampled from SFT model. Reward Function: human preference in the general domain (can be 'Rule' in mathematical tasks). Gradient Coefficient:

$$GC_{DPO}(q, o, t) = \sigma \left(\beta \log \frac{\pi_{\theta}(o_t^-|q, o_{<t}^-)}{\pi_{ref}(o_t^-|q, o_{<t}^-)} - \beta \log \frac{\pi_{\theta}(o_t^+|q, o_{<t}^+)}{\pi_{ref}(o_t^+|q, o_{<t}^+)} \right) \quad (14)$$

A.1.5. Proximal Policy Optimization (PPO)

The objective of PPO is:

$$\mathcal{J}_{PPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o \sim \pi_{\theta_{old}}(O|q)] \frac{1}{|o|} \sum_{t=1}^{|o|} \min \left[\frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})} A_t, \text{clip} \left(\frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})}, 1 - \epsilon, 1 + \epsilon \right) A_t \right]. \quad (15)$$

To simplify the analysis, it is assumed that the model only has a single update following each exploration stage, thereby ensuring that $\pi_{\theta_{old}} = \pi_{\theta}$. In this case, we can remove the min and clip operation:

$$\mathcal{J}_{PPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o \sim \pi_{\theta_{old}}(O|q)] \frac{1}{|o|} \sum_{t=1}^{|o|} \frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})} A_t. \quad (16)$$

The gradient of $\mathcal{J}_{PPO}(\theta)$ is:

$$\nabla_{\theta} \mathcal{J}_{PPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o \sim \pi_{\theta_{old}}(O|q)] \frac{1}{|o|} \sum_{t=1}^{|o|} A_t \nabla_{\theta} \log \pi_{\theta}(o_t|q, o_{<t}) \quad (17)$$

Data Source: question in SFT dataset with outputs sampled from policy model. Reward Function: reward model. Gradient Coefficient:

$$GC_{PPO}(q, o, t, \pi_{\theta_{rm}}) = A_t, \quad (18)$$

where A_t is the advantage, which is computed by applying Generalized Advantage Estimation (GAE) (Schulman et al., 2015), based on the rewards $\{r_{\geq t}\}$ and a learned value function V_{ψ} .

A.1.6. Group Relative Policy Optimization (GRPO)

The objective of GRPO is (assume $\pi_{\theta_{old}} = \pi_{\theta}$ for simplified analysis):

$$\begin{aligned} \mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)] \\ \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left[\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t} - \beta \left(\frac{\pi_{ref}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} - \log \frac{\pi_{ref}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} - 1 \right) \right]. \end{aligned} \quad (19)$$

The gradient of $\mathcal{J}_{GRPO}(\theta)$ is:

$$\begin{aligned} \nabla_{\theta} \mathcal{J}_{GRPO}(\theta) &= \mathbb{E}[q \sim P_{sft}(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)] \\ &\quad \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left[\hat{A}_{i,t} + \beta \left(\frac{\pi_{ref}(o_{i,t}|o_{i,<t})}{\pi_{\theta}(o_{i,t}|o_{i,<t})} - 1 \right) \right] \nabla_{\theta} \log \pi_{\theta}(o_{i,t}|q, o_{i,<t}). \end{aligned} \quad (20)$$

Data Source: question in SFT dataset with outputs sampled from policy model. Reward Function: reward model. Gradient Coefficient:

$$GC_{GRPO}(q, o, t, \pi_{\theta_{rm}}) = \hat{A}_{i,t} + \beta \left(\frac{\pi_{ref}(o_{i,t}|o_{i,<t})}{\pi_{\theta}(o_{i,t}|o_{i,<t})} - 1 \right), \quad (21)$$

where $\hat{A}_{i,t}$ is computed based on the group reward scores.